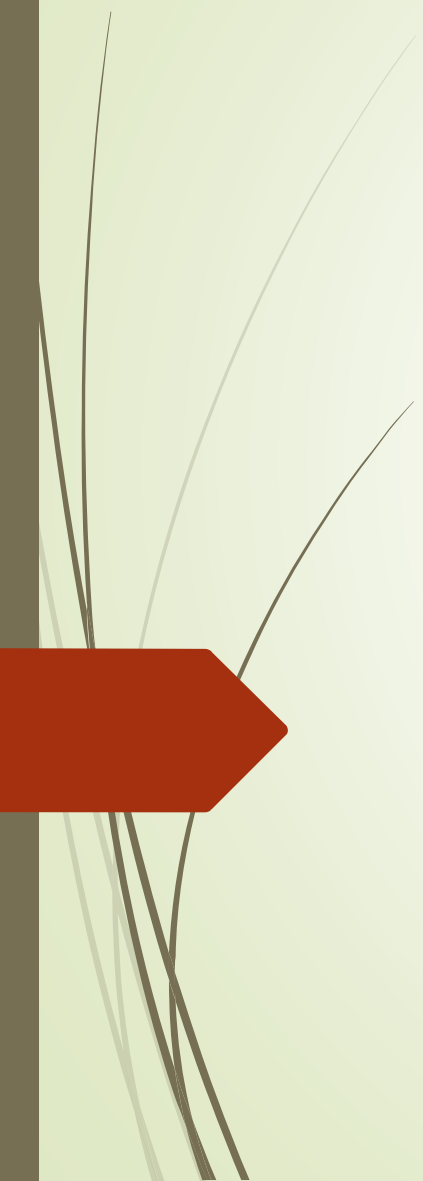




Basic Network Management



Dynamic Host Configuration Protocol (DHCP) & DHCP Server

- Dynamic Host Configuration Protocol (DHCP) is a client/server protocol that automatically provides an Internet Protocol (IP) host with its IP address and other related configuration information such as the subnet mask and default gateway
- A DHCP Server is **a network server that automatically provides and assigns IP addresses, default gateways and other network parameters to client devices.** It relies on the standard protocol known as Dynamic Host Configuration Protocol or DHCP to respond to broadcast queries by clients

Why use DHCP?

- Every device on a TCP/IP-based network must have a unique unicast IP address to access the network and its resources. Without DHCP, IP addresses for new computers or computers that are moved from one subnet to another must be configured manually; IP addresses for computers that are removed from the network must be manually reclaimed.
- With DHCP, this entire process is automated and managed centrally. The DHCP server maintains a pool of IP addresses and leases an address to any DHCP-enabled client when it starts up on the network. Because the IP addresses are dynamic (leased) rather than static (permanently assigned), addresses no longer in use are automatically returned to the pool for reallocation.
- The network administrator establishes DHCP servers that maintain TCP/IP configuration information and provide address configuration to DHCP-enabled clients in the form of a lease offer. The DHCP server stores the configuration information in a database that includes:
 - Valid TCP/IP configuration parameters for all clients on the network.

Why use DHCP?

- Valid IP addresses, maintained in a pool for assignment to clients, as well as excluded addresses.
- Reserved IP addresses associated with particular DHCP clients. This allows consistent assignment of a single IP address to a single DHCP client.
- The lease duration, or the length of time for which the IP address can be used before a lease renewal is required.
- A DHCP-enabled client, upon accepting a lease offer, receives:
- A valid IP address for the subnet to which it is connecting.
- Requested DHCP options, which are additional parameters that a DHCP server is configured to assign to clients. Some examples of DHCP options are Router (default gateway), DNS Servers, and DNS Domain Name.



Benefits of DHCP

- DHCP provides the following benefits.
- **Reliable IP address configuration.** DHCP minimizes configuration errors caused by manual IP address configuration, such as typographical errors, or address conflicts caused by the assignment of an IP address to more than one computer at the same time.
- **Reduced network administration.** DHCP includes the following features to reduce network administration:
 - Centralized and automated TCP/IP configuration.
 - The ability to define TCP/IP configurations from a central location.
 - The ability to assign a full range of additional TCP/IP configuration values by means of DHCP options.
 - The efficient handling of IP address changes for clients that must be updated frequently, such as those for portable devices that move to different locations on a wireless network.
 - The forwarding of initial DHCP messages by using a DHCP relay agent, which eliminates the need for a DHCP server on every subnet.

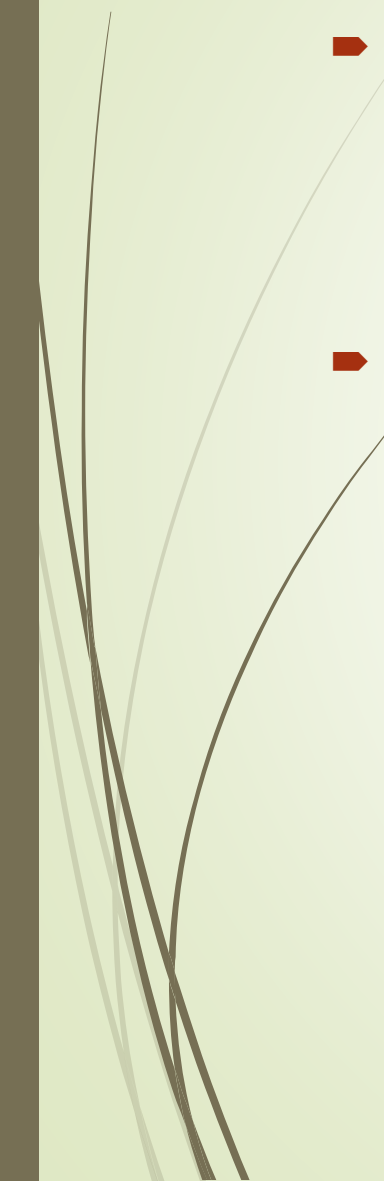


DHCP Server Configuration

- <https://www.redhat.com/sysadmin/static-dynamic-ip-2>



NIS & NFS

- The Network Information Service (NIS) and Network File System (NFS) are services that allow you to build distributed computing systems that are both consistent in their appearance and transparent in the way files and data are shared.
 - NFS and NIS are high-level networking protocols, built on several lower-level protocols.
- 



Network Information Service (NIS)

- NIS provides a distributed database system for common configuration files.
- NIS servers manage copies of the database files, and NIS clients request information from the servers instead of using their own, local copies of these files. For example, the */etc/hosts* file is managed by NIS.
- A few NIS servers manage copies of the information in the hosts file, and all NIS clients ask these servers for host address information instead of looking in their own */etc/hosts* file.
- Once NIS is running, it is no longer necessary to manage every */etc/hosts* file on every machine in the network — simply updating the NIS servers ensures that all machines will be able to retrieve the new configuration file information.



Network File System (NFS)

- NFS is a distributed file system.
- An NFS server has one or more file systems that are mounted by NFS clients; to the NFS clients, the remote disks look like local disks.
- NFS file systems are mounted using the standard Unix *mount* command, and all Unix utilities work just as well with NFS-mounted files as they do with files on local disks.
- NFS makes system administration easier because it eliminates the need to maintain multiple copies of files on several machines: all NFS clients share a single copy of the file on the NFS server.
- NFS also makes life easier for users: instead of logging on to many different systems and moving files from one system to another, a user can stay on one system and access all the files that he or she needs within one consistent file tree.

SAMBA

- **Samba is the standard Windows interoperability suite of programs for Linux and Unix.**
- A Samba file server enables file sharing across different operating systems over a network. It lets you access your desktop files from a laptop and share files with Windows and mac OS users.
- Samba is a suite of applications that implements the Server Message Block (SMB) protocol. Many operating systems, including Microsoft Windows, use the SMB protocol for client-server networking. Samba enables Linux / Unix machines to communicate with Windows machines in a network.
- Samba is open source software. Originally, Samba was developed in 1991 for fast and secure file and print share for all clients using the SMB protocol. Since then it has evolved and added more capabilities. Today Samba provides a suite of applications enabling seamless networking and interoperability between *nix and Windows.
- <https://ubuntu.com/tutorials/install-and-configure-samba#1-overview>

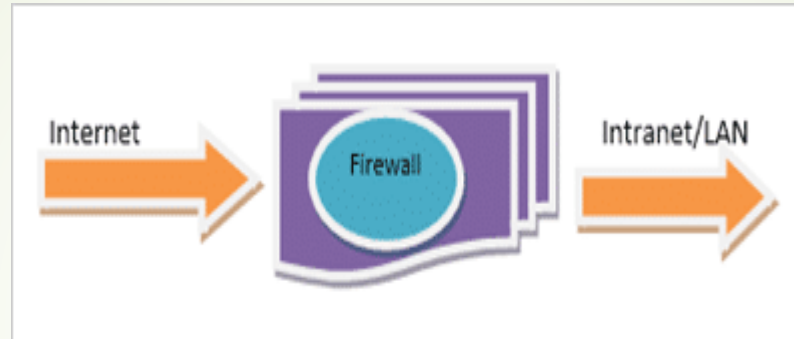


Firewall

- The concept of the firewall was introduced to secure the communication process between various networks.
- A firewall is a software or a hardware device that examines the data from several networks and then either permits it or blocks it to communicate with your network and this process is governed by a set of predefined security guidelines.
- A firewall is a device or a combination of systems that supervises the flow of traffic between distinctive parts of the network. A firewall is used to guard the network against nasty people and prohibit their actions at predefined boundary levels.
- A firewall is not only used to protect the system from exterior threats but the threat can be internal as well. Therefore we need protection at each level of the hierarchy of networking systems.
- A good firewall should be sufficient enough to deal with both internal and external threats and be able to deal with malicious software such as worms from acquiring access to the network. It also provisions your system to stop forwarding unlawful data to another system.

Firewall

- a firewall always exists between a private network and the Internet which is a public network thus filters packets coming in and out.



Selecting a precise firewall is critical in building up a secure networking system.

Firewall provisions the security apparatus for allowing and restricting traffic, authentication, address translation, and content security.

It ensures 365 *24*7 protection of the network from hackers. It is a one-time investment for any organization and only needs timely updates to function properly.

By deploying a firewall there is no need for any panic in case of network attacks.

How does a firewall work?

- A firewall establishes a border between an external network and the network it guards. It is inserted inline across a network connection and inspects all packets entering and leaving the guarded network. As it inspects, it uses a set of pre-configured rules to distinguish between benign and malicious packets.
- The term 'packets' refers to pieces of data that are formatted for internet transfer. Packets contain the data itself, as well as information about the data, such as where it came from. Firewalls can use this packet information to determine whether a given packet abides by the rule set. If it does not, the packet will be barred from entering the guarded network.
- Rule sets can be based on several things indicated by packet data, including:

Their source.

Their destination.

Their content.



Types of Firewall

➤ 1. Software Firewall :

Software firewall is a special type of computer software runs on a computer/server. It's main purpose is to protect your computer/server from outside attempts to control or gain access and depending on your choice of software firewall. Software firewall can also be configured for checking any suspicious outgoing requests.

➤ 2. Hardware Firewall :

It is physical piece of equipment planned to perform firewall duties. A hardware firewall can be a computer or a dedicated piece of equipment which serve as a firewall. Hardware firewall are incorporated into the router that is situated between the computer and the internet gateway.

Types of Firewall

■ Packet Filtering Firewalls

Packet filtering firewalls are the oldest, most basic type of firewalls. Operating at the network layer, they check a data packet for its source IP and destination IP, the protocol, source port, and destination port against predefined rules to determine whether to pass or discard the packet.

■ Circuit-Level Gateways

Working at the session layer, circuit-level gateways verify established Transmission Control Protocol (TCP) connections and keep track of the active sessions. Primarily, they determine the security of an established connection. When an internal device initiates a connection with a remote host, circuit-level gateways establish a virtual connection on behalf of the internal device to keep the identity and IP address of the internal user hidden.

■ Stateful Inspection Firewalls

stateful inspection firewalls, and verifying and keeping track of established connections also perform packet inspection to provide better, more comprehensive security. They work by creating a state table with source IP, destination IP, source port, and destination port once a connection is established. They create their own rules dynamically to allow expected incoming network traffic instead of relying on a hardcoded set of rules based on this information. They conveniently drop data packets that do not belong to a verified active connection.

■ Application-Level Gateways (Proxy Firewalls)

Application-level gateways, also known as proxy firewalls, are implemented at the application layer via a proxy device. Instead of an outsider accessing your internal network directly, the connection is established through the proxy firewall. The external client sends a request to the proxy firewall. After verifying the authenticity of the request, the proxy firewall forwards it to one of the internal devices or servers on the client's behalf. Alternatively, an internal device may request access to a webpage, and the proxy device will forward the request while hiding the identity and location of the internal devices and network.

Designing a Firewall

- iptables - administration tool for IPv4 packet filtering and NAT
- **Iptables** is used to set up, maintain, and inspect the tables of IP packet filter rules in the Linux kernel. Several different tables may be defined. Each table contains a number of built-in chains and may also contain user-defined chains.
- Each chain is a list of rules which can match a set of packets. Each rule specifies what to do with a packet that matches. This is called a 'target', which may be a jump to a user-defined chain in the same table.
- A firewall rule specifies criteria for a packet, and a target. If the packet does not match, the next rule in the chain is the examined; if it does match, then the next rule is specified by the value of the target, which can be the name of a user-defined chain or one of the special values *ACCEPT*, *DROP*, *QUEUE*, or *RETURN*.
- *ACCEPT* means to let the packet through. *DROP* means to drop the packet on the floor. *QUEUE* means to pass the packet to userspace. *RETURN* means stop traversing this chain and resume at the next rule in the previous (calling) chain.
- <https://linux.die.net/man/8/iptables>
- <https://www.youtube.com/watch?v=vbhr4csDeI4>
- <https://www.youtube.com/watch?v=6Ra17Qpj68c>